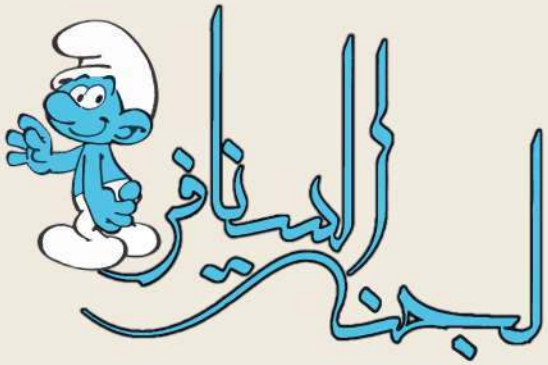
خدمتكم طريق خضناه لرضى الله

2021

# أسئلة سنوات فاينال

# كيمياء عامة





# بسم الله الرحمن الرحيم

نقدم لكم نحن أسرة فريق

(لجنة السنافر)

مجموعة أسئلة اختبارات إلكترونية تم تجميعها خلال الفصول الماضية
سائلين المولى أن يوفقنا وإياكم لكل خير

## تنويه

يوجد بعض الأسئلة عليها إجابات قد تحتمل الصواب وقد تحتمل الخطأ

فإن أصبنا فما هو إلا توفيق من الله

وإن أخطأنا فمن أنفسنا

#خدمتكم_طريق_خضناه_لرضى_الله

#الإتجاه_الاسلامي

#بسواعدنا_نبنيها

#لجنة_السنافر

#هي_لله

The boiling point of a substance X = 350 K

If enthalpy of vaporization ($\Delta H$ vap) = 55kJ/mol

Find the entropy of vaporization ($\Delta S$ vap) in J/mol.K

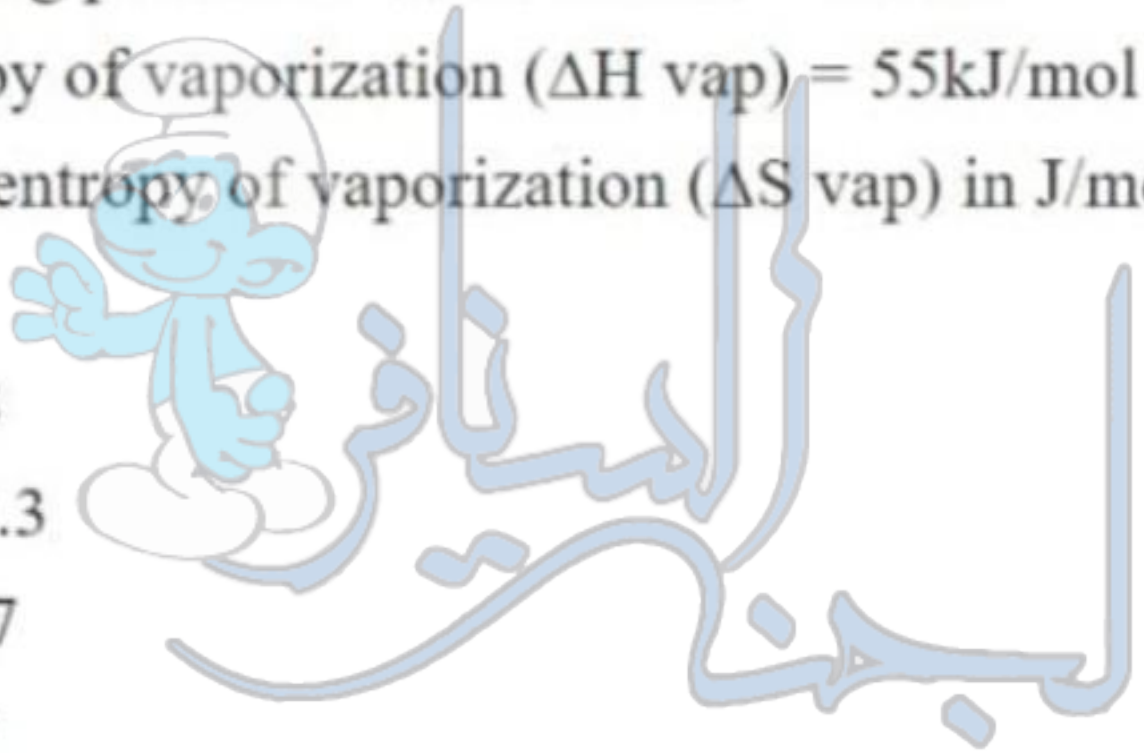
A. 298

B. 636.3

C. -157

D. 157

E. -636.3



Which of the following is correct regarding bomb calorimetry and coffee-cup calorimetry

Select one:

- In bomb calorimetry the volume remains constant
- in bomb calorimetry work is zero
- Coffee cup calorimeter is mostly used for solution reactions
- In both techniques, heat flow is measured through temperature changes
- All are correct

If piece of metal at 120 °C is placed in 106 g of $H_2O$ at 20 °C.

The final temperature of both water an metal piece was 25 °C.

Calculate heat lost by metal in J

(Specific heat for $H_2O$ = 4.184 $J.g^{-1}.C^{-1}$ )

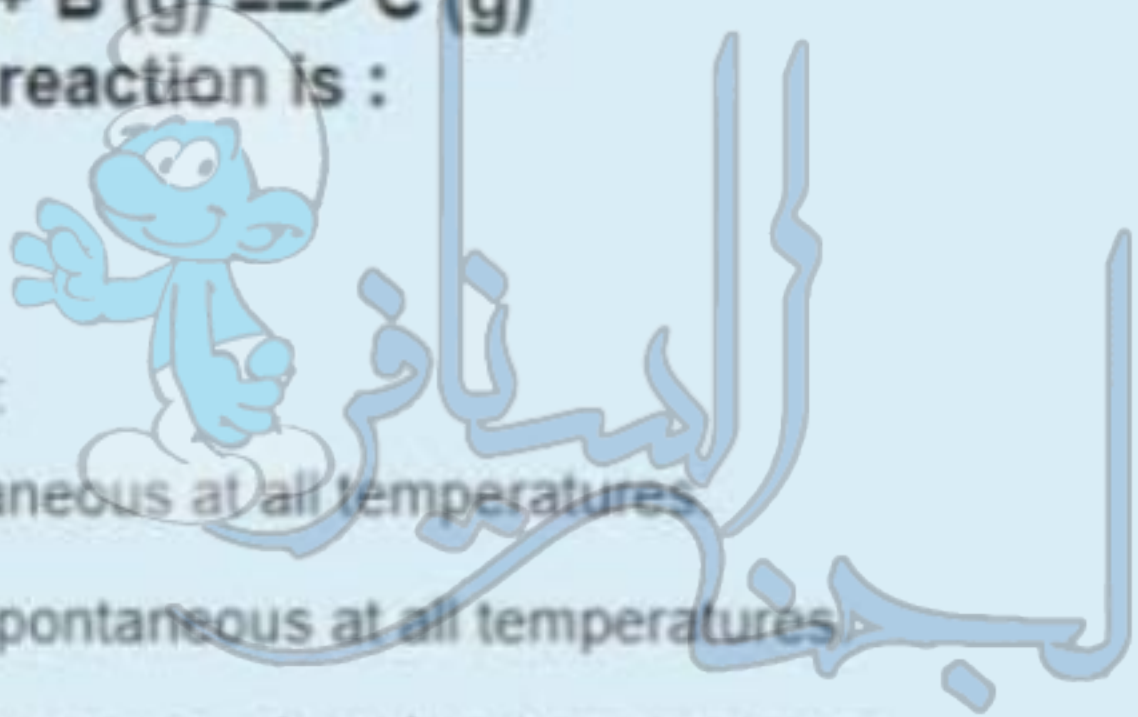
Select one:

- 2217.5
- -1774
- 3322.4
- 1774
- -2217.5

For the following exothermic reaction

$A(g) + B(g) \Longrightarrow C(g)$

This reaction is :



Select one:

- spontaneous at all temperatures.
- non-spontaneous at all temperatures.
- spontaneous only at low temperatures.
- spontaneous only at high temperatures.

Given the following data for this reaction:

$$NH_4^+(aq) + NO_2^-(aq) \Longrightarrow N_2(g) + 2H_2O(l)$$

| EXP | $[NH_4^+]$ | $[NO_2^-]$ | RATE |
|---|---|---|---|
| 1 | 0.010 M | 0.020 M | 0.020 M/s |
| 2 | 0.015 M | 0.020 M | 0.030 M/s |
| 3 | 0.010 M | 0.010 M | 0.005 M/s |

Select one:

- Rate = $k[NH_4^+][NO_2^-]$
- Rate = $k[NH_4^+]^2[NO_2^-]^2$
- Rate = $k[NH_4^+]^2[NO_2^-]$
- Rate = $k[NH_4^+][NO_2^-]^2$
- none

**Calculate percentage by mass concentration for a 0.82 M $H_3PO_4$ solution (molar mass =98 g/mol) if density of solution = 1.12 g/mol**

Select one:

- 7.2 %
- 12.4%
- 6.3 %
- 10.2%
- 8.05%

The equation for the standard enthalpy of formation of calcium carbonate ($CaCO_3$) is:

A. $Ca(s) + C(s) + 3/2\ O_2(l) \rightarrow CaCO_3(s)$

B. $Ca(s) + 2C(s) + 3/2\ O_2(g) \rightarrow CaCO_3(s)$

C. $Ca(s) + C(s) + 3/2\ O_2(g) \rightarrow CaCO_3(s)$

D. $Ca(s) + C(s) + 3O(g) \rightarrow CaCO_3(s)$

Select one:

- A
- B
- C
- D

When two solutions react the container "feels cold" That means that:

A) the reaction is endothermic.

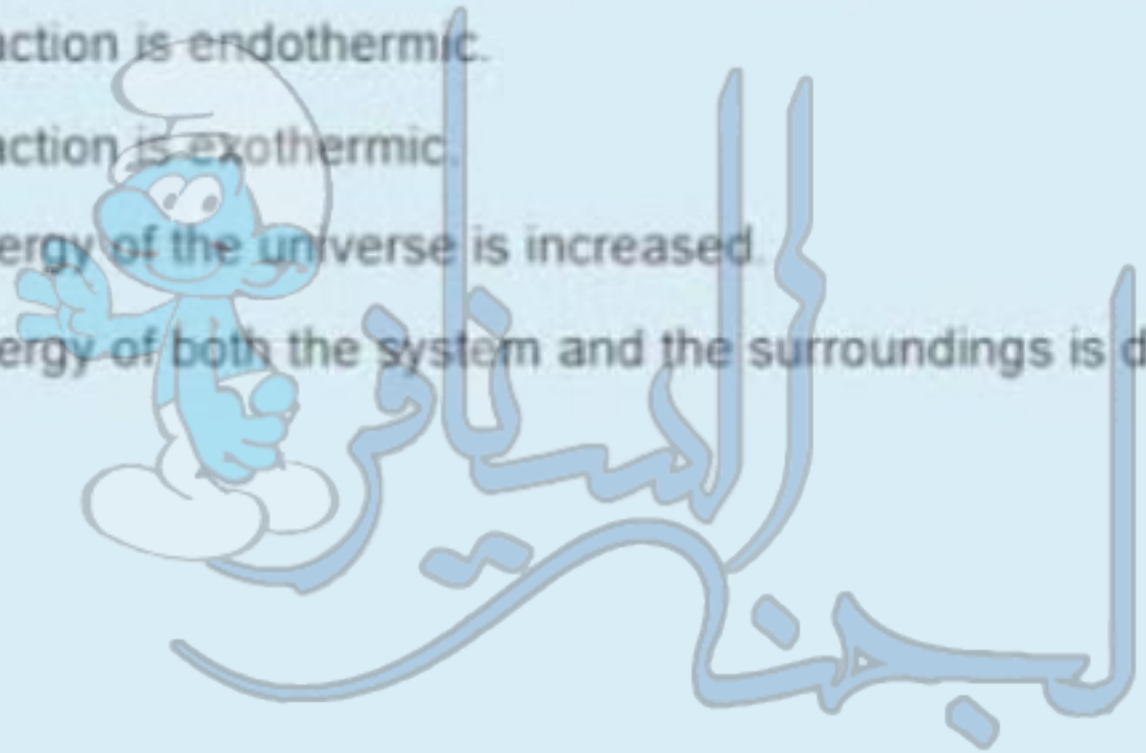
B) the reaction is exothermic.

C) the energy of the universe is increased.

D) the energy of both the system and the surroundings is decreased.

Select one:

- A
- B
- C
- D



For the given reaction :

**$2A + 3B + \frac{1}{2}C \rightarrow 4D + 2E$**

If 4 mol (**C**) absorbs 400 kJ of heat (q)

Calculate ΔH for the above reaction (in kJ)



a. +25

b. +50

c. -25

d. -50

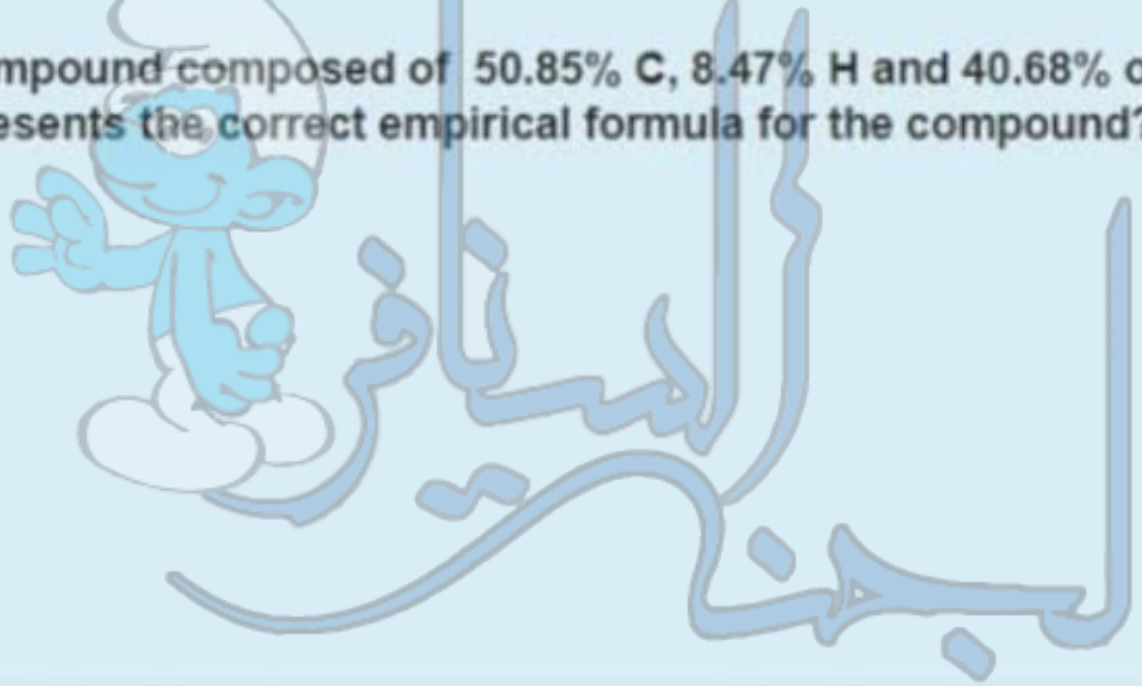
Select one:

- a
- b
- c
- d

**An organic compound composed of 50.85% C, 8.47% H and 40.68% oxygen. Which of the following represents the correct empirical formula for the compound?**

Select one:

- $C_5H_{10}O_3$
- $C_3H_6O_2$
- $CH_2O$
- $C_2H_4O$
- $C_4H_8O_3$



You can find 2 moles of oxygen atoms in:

(a) 0.5 mole of $K_2SO_4$
(b) 4 moles of $Na_2O$
(c) 1 molecule of $Na_3PO_4$
(d) 4 molecules of $H_2O$
(e) 2 grams of $Ca(OH)_2$ (76g/mol)

(Avog.no $=6.022\times10^{23}$ )

Select one:

- a
- b
- c
- d
- e

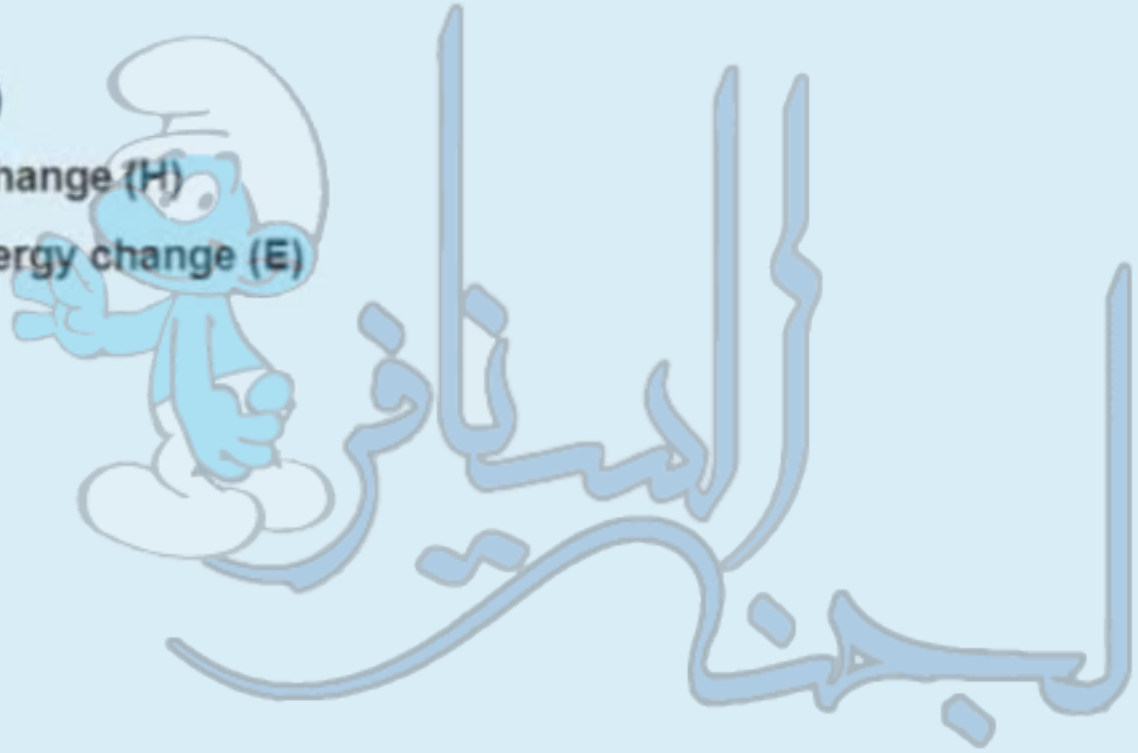
Which of the following thermodynamic quantities are not state functions?

Heat (q)

Entropy (S)

Enthalpy change (H)

Internal energy change (E)



Select one:

- q only
- E and S
- H and E
- q and E
- H and q

**Calculate work accompanied to expansion of a 4L gas inside a cylinder against external atmospheric pressure = 250 kPa until the volume of gas became 6L.**

Select one:

- -1000 J
- +500 J
- -500 J
- +1000 J
- -1500 J

Calculate $\Delta G^\circ$ rxn : $C_3H_6O\ (l) + 4O_2\ (g) \rightarrow 3H_2O\ (l) + 3CO_2(g)$

given that:

$\Delta G_f^\circ\ CO_2(g) = -394.6$ kJ/mol,

$\Delta G_f^\circ\ H_2O(l) = -237.2$ kJ/mol and

$\Delta G^\circ f\ C_3H_6O\ (l) = -154.5$ kJ/mol,



Select one:

- +1740.9 kJ
- -1740.9 KJ
- -3481.8 KJ
- +3481.8 KJ
- -3697.8 KJ

The half-life for a first-order reaction is 70 s.

What was the original concentration if, after 2.0 minutes, the reactant concentration remained is 0.062 M?

Select one:

- 0.421M
- 0.204 M
- 0.368 M
- 0.0189 M
- 0.668 M

What is the molar concentration of a solution of 9.88 g $MgSO_4$ dissolved in 75 mL water ?

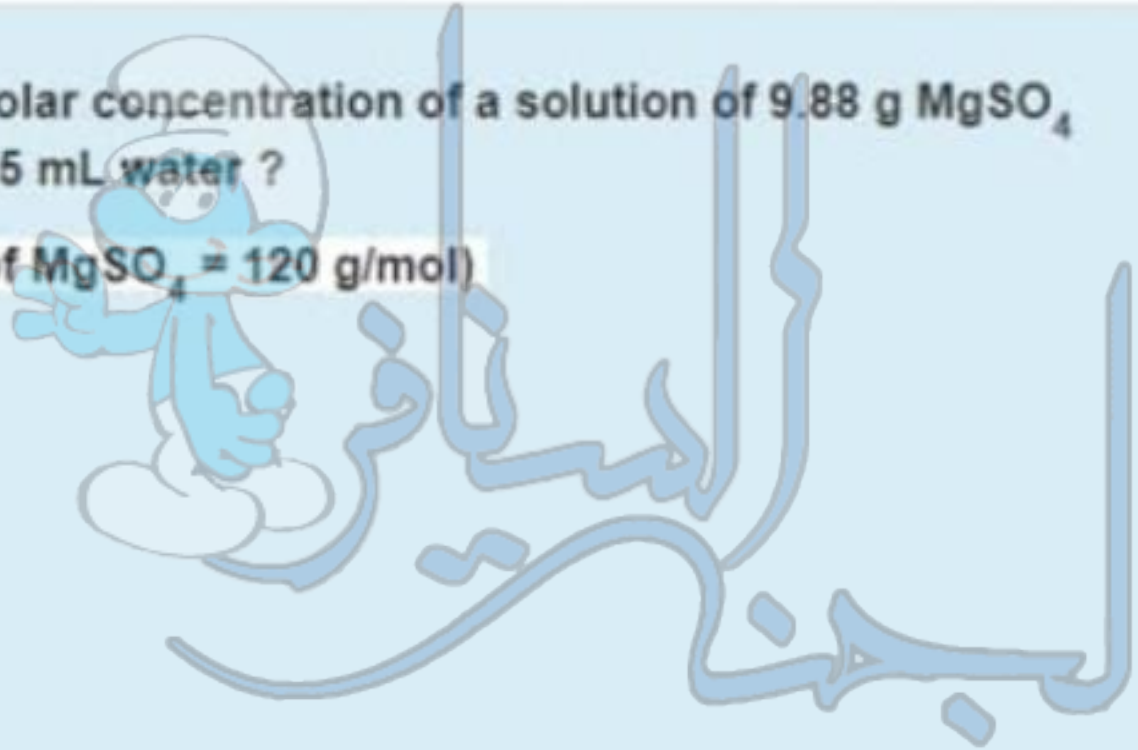
(Molar mass of $MgSO_4$ = 120 g/mol)

a. 0.011 M

b. 11 M

c. 1.1 M

d. 0.11 M



Calculate $\Delta H$ for the reaction

$IF_{5\,(g)} \rightarrow IF_{3\,(g)} + F_{2\,(g)}$ is _____ kJ,

given the data below.

$IF_{(g)} + F_{2\,(g)} \rightarrow IF_{3\,(g)}$ $\Delta H = -390$ kJ

$1/2\ IF_{(g)} + F_{2\,(g)} \rightarrow 1/2\ IF_{5\,(g)}$ $\Delta H = -372.5$ kJ

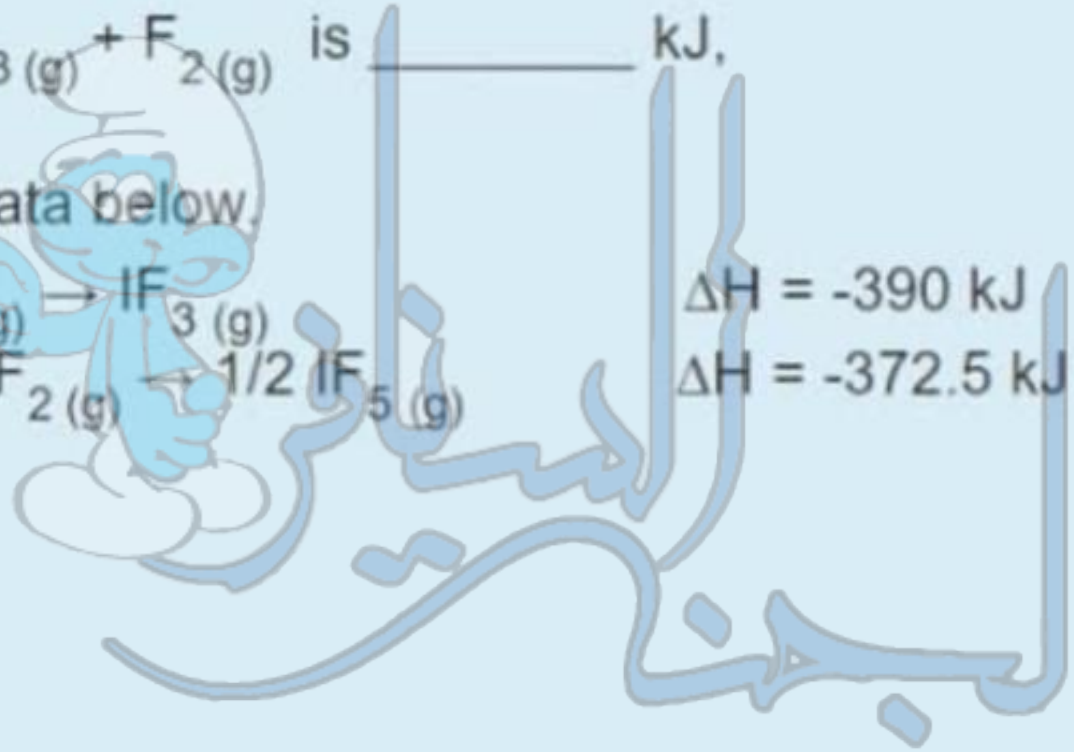
A) -1135

B) +35

C) +1135

D) -35

E) +355

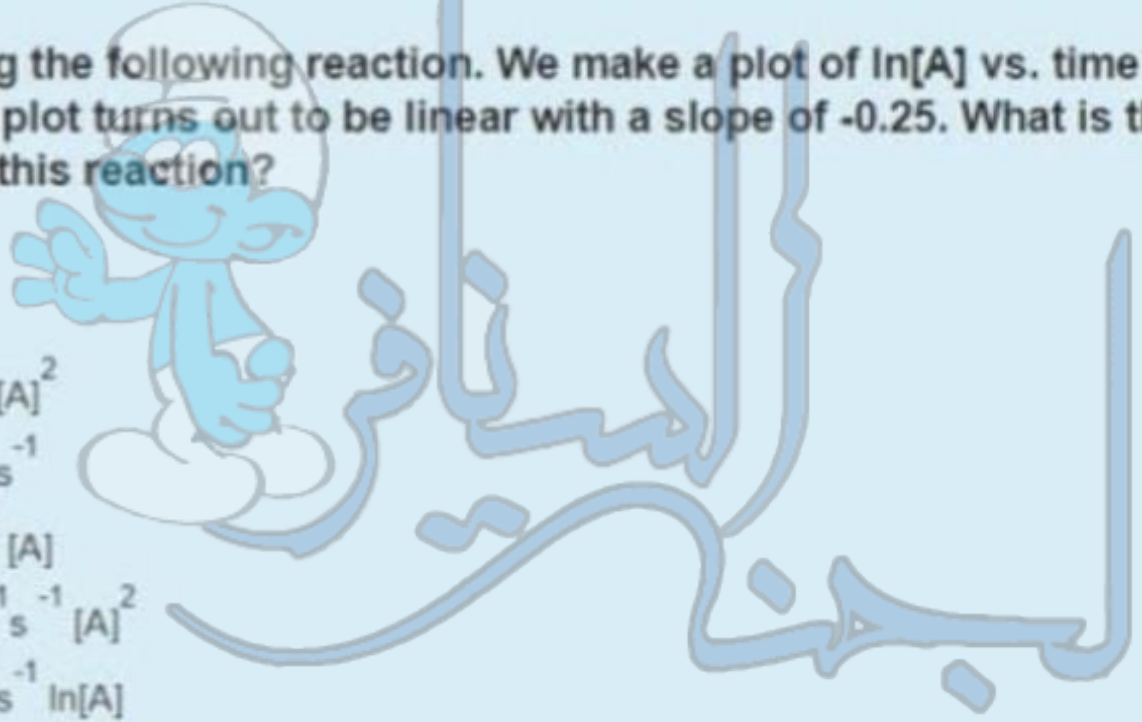


We are studying the following reaction. We make a plot of ln[A] vs. time as the reaction proceeds. This plot turns out to be linear with a slope of -0.25. What is the rate law expression for this reaction?

$2\,A \rightarrow 3\,B$

Select one:

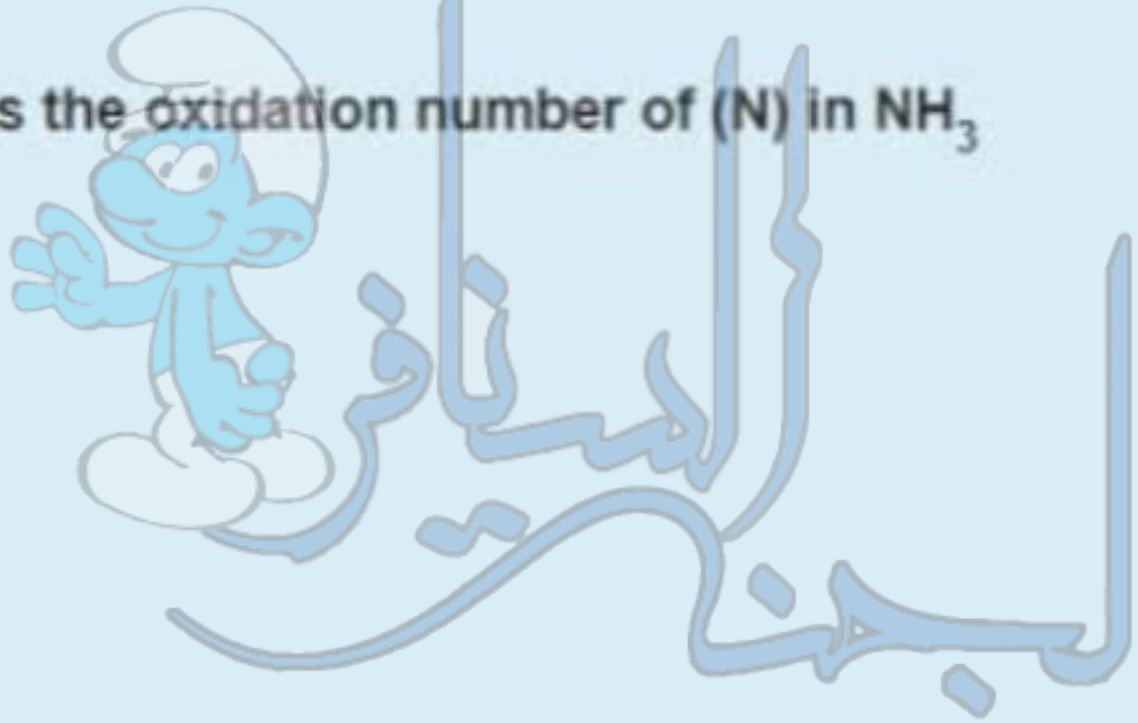
- ate = $0.25\ s^{-1}\ [A]^2$
- rate = $0.25\ M.s^{-1}$
- rate = $0.25\ s^{-1}\ [A]$
- rate = $0.25\ M^{-1} s^{-1}\ [A]^2$
- rate = $0.25\ M.s^{-1}\ \ln[A]$



What is the oxidation number of (N) in $NH_3$

Select one:

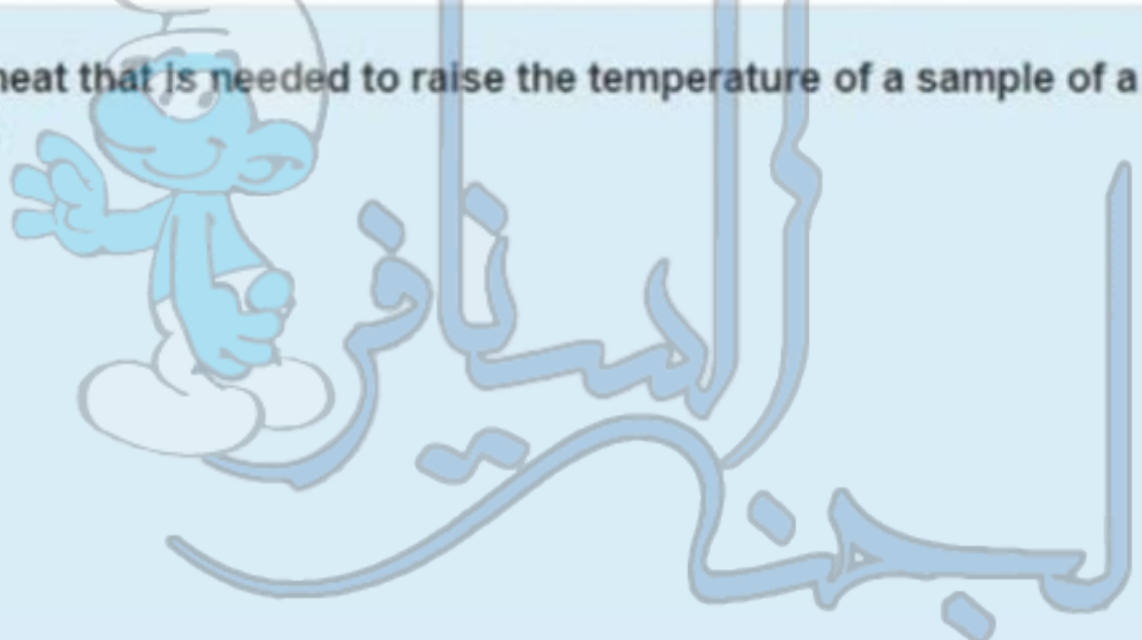
- -3
- +4
- +3
- -4
- +5



**The quantity of heat that is needed to raise the temperature of a sample of a substance 1.00 degree is called:**

Select one:

- Heat capacity
- Specific heat
- Kinetic energy
- Enthalpy



A representative element in period 4 has only one electron in its valence shell (outer shell) what is the atomic number of this element

Select one:

- 21
- 13
- 12
- 20
- 19

Rate constants for the first-order reaction:

rate constant $k = 4.75 \times 10^{-4}\ s^{-1}$ at 293 K

rate constant $k = 1.63 \times 10^{-3}$ at 310 K.

What is the activation energy, $E_a$ for this reaction?

R =8.314 J/mol.K

Select one:

- 91 kJ
- -91.1 kJ
- 35.6 kJ
- 54.8 kJ
- 44.7 kJ

For the reaction:

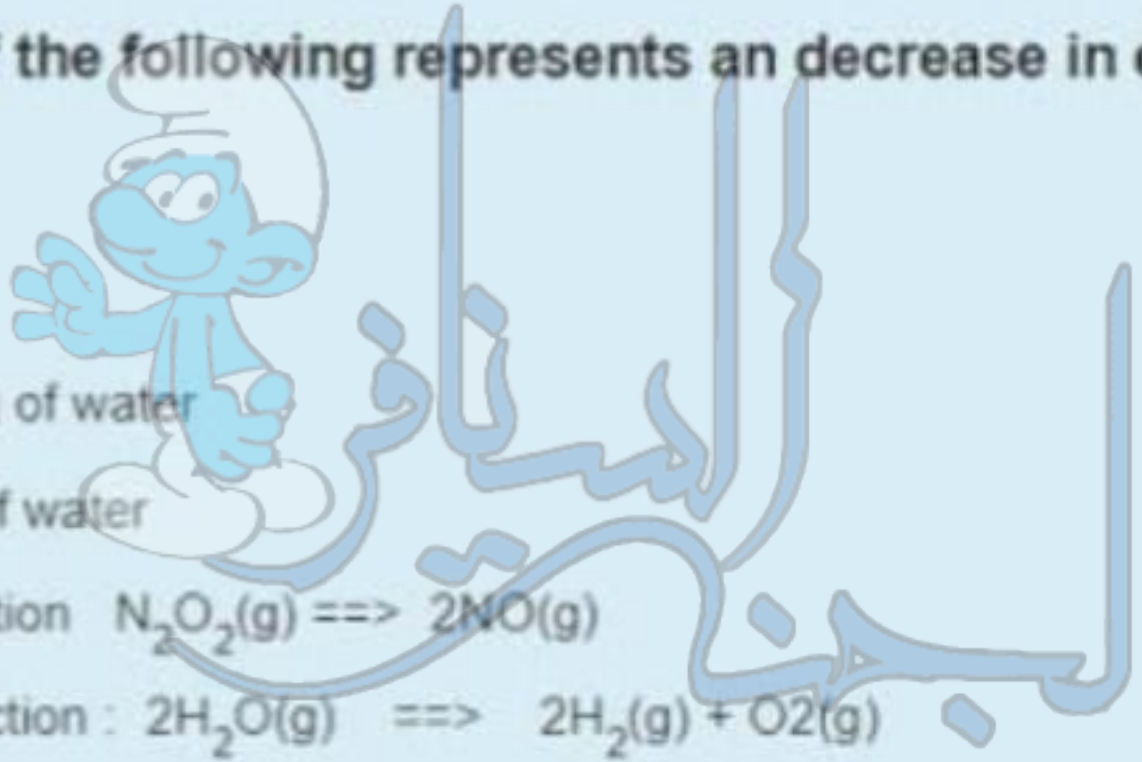
$$2H_3(PO_4)_{2(aq)} + 3Ca(OH)_{2(aq)} \rightarrow Ca_3(PO_4)_{2(s)} + 6H_2O(l)$$

Calculate mass of $Ca_3(PO_4)_2$(molar mass =310 g//mol) precipitated when 6 mole of $H_3(PO_4)_2$ are allowed to react with 12 mole of $Ca(OH)_2$.

Select one:

- 1550 g
- 1240 g
- 310 g
- 930 g
- 620 g

**Which of the following represents an decrease in entropy?**



Select one:

- freezing of water
- boiling of water
- the reaction $N_2O_2(g) \Longrightarrow 2NO(g)$
- The reaction : $2H_2O(g) \Longrightarrow 2H_2(g) + O2(g)$
- none

When the following equation is balanced with the smallest possible set of integers (in the acidic medium)

$$Mn^{2+}(aq) + Fe^{3+}(aq) \Longrightarrow Fe^{2+}(aq) + MnO_4^-(aq)$$

The coefficient of $Fe^{3+}$ in the balanced equation is :

Select one:

- 5
- 2
- 8
- 1
- 4

?Which has the highest first ionization energy



$_{11}A$ $_{12}B$ $_{13}C$

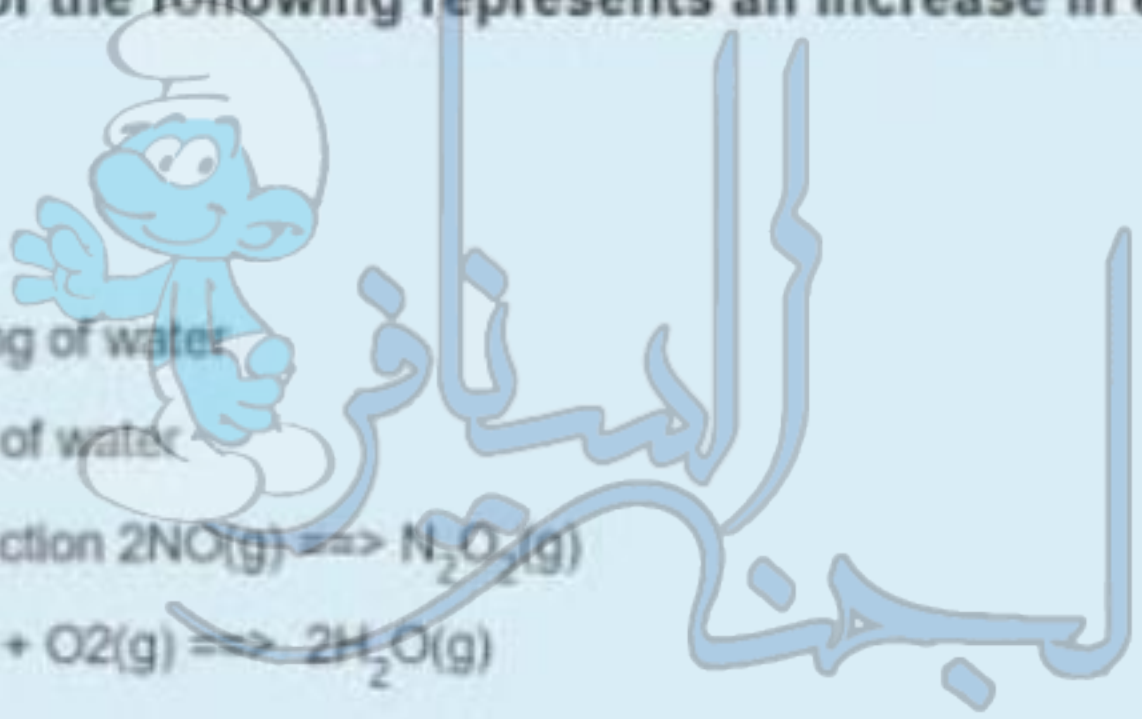
اختر أحد الخيارات

C ✓

A

B

**Which of the following represents an increase in entropy?**



Select one:

- freezing of water
- boiling of water
- the reaction $2NO(g) \Longrightarrow N_2O_2(g)$
- $2H_2(g) + O2(g) \Longrightarrow 2H_2O(g)$
- none

**An organic compound composed of 48.65 % C, 8.11% H and 43.24 % oxygen. Which of the following represents the correct empirical formula for the compound?**

Select one:

- $C_4H_8O_3$
- $C_3H_6O_2$
- $C_2H_4O$
- $C_5H_{10}O_3$
- $CH_2O$